

MEVIN JOSE

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Work Experience

Wadhvani Foundation

Bangalore, India

AI Intern

Jun 2026 – Aug 2026

- Worked on Content Ingestion for the backend RAG system of **Wadhvani Liftoff.ai**, an enterprise AI platform, studying document processing, embeddings, vector search, and retrieval architecture.
- Traced Python/FastAPI workflows spanning OCR, LLM extraction, multilingual processing, embeddings, and Qdrant-based semantic retrieval.
- Studied embedding-model migration and source-grounded RAG across the content-ingestion and retrieval lifecycle.

Superhhero Learning

Bangalore, India

Web Developer Intern

Jun 2025 – Sep 2025

- Built **4 of 5 application routes** and **5 core UI modules** for an online education platform using **Next.js, React, and Tailwind CSS**, delivering responsive, interactive, data-driven components through **8 merged PRs**.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java

ML & AI: PyTorch, Scikit-learn, LightGBM, OpenCV, SHAP, NLP

LLM & Evaluation: RAG, FAISS, Sentence Transformers, LangChain, LangGraph, Text-to-SQL, SQLGlot, AST Verification, Benchmarking, Ablation Studies, RAGAS, PEFT (LoRA)

MLOps & Infrastructure: FastAPI, Docker, GitHub Actions, AWS (EC2, S3), Drift Detection, Model Monitoring, CI/CD

Databases & Tools: PostgreSQL, MongoDB, Redis, SQLite, Qdrant, Pinecone, Pandas, NumPy, SciPy

Projects

Engineering Reliable LLM-Based Data Analysis – Research Preprint ^{DOI}

Python, SQL, RAG, SQLGlot

- Built a reliability-focused LLM system that converts business questions into validated, executable SQL while detecting schema, join, and aggregation errors.
- Combined graph-guided schema RAG, DAG planning, deterministic validation, AST-based SQL verification, and closed-loop repair.
- Evaluated **500 queries** across **8 e-commerce domains**, achieving **73.4% result equivalence**, **31.0% exact match**, and **100% SQL execution success**.
- Conducted a controlled **100-query** ablation showing AST verification improved result equivalence from **15.0% to 26.0%** ($p = 0.0192$, $OR=3.75$); audited **101** repair events.

SentinelGraph – Explainable Anomaly Detection Engine

Python, Streamlit, NetworkX, PyTorch, Docker

- Built an explainable security platform that detects anomalous enterprise access behavior using behavioral, statistical, temporal, graph, and sequence signals.
- Architected a **7-detector** anomaly ensemble for **1,000+ entities**, suppressing false-positive alerts by **>90%** through max-dominant risk fusion.
- Achieved **0.942 PR-AUC** and **sub-10ms** scoring latency over **10K benchmark events** using GNN and Graph-RAG traversal pipelines.
- Integrated counterfactual explainability and SHAP feature breakdowns, increasing threat-triage efficiency by **30%**.

UnderdogAI – High-Throughput Event-Driven Decision Platform

FastAPI, Kafka, gRPC, Redis, PostgreSQL, Kubernetes

Kubernetes

- Built a decision-intelligence platform that predicts football upset probabilities and runs large-scale tournament simulations through distributed backend services.
- Built fault-tolerant FastAPI/Kafka/gRPC services with asynchronous Monte Carlo workers, Redis task state, PostgreSQL storage, and Kubernetes deployments.
- Increased API throughput **57%** from 352 to 553 req/s at **50 concurrent clients**, reducing P50/P95/P99 latency by **37%/26%/29%** with zero failures.
- Reduced server CPU time **42%** and peak RSS **10%**; recovered two injected failures in **3.0s**.

Achievements

3× DAC Scholarship Recipient, PES University; **350+ LeetCode problems solved**; **Top 10, Kodikon National-Level Hackathon** among **280+ teams**; **Technical Team Member, The Alcoding Club**.

Education

PES University

Bangalore, India

B.Tech, Computer Science (AI & Machine Learning)

Aug 2023 – May 2027

CGPA: 7.85

Coursework: Explainable AI, Large Language Models and Their Applications, Advanced Foundations for Machine Learning, Machine Learning, Deep Learning, Reinforcement Learning, Optimizations and Metaheuristics